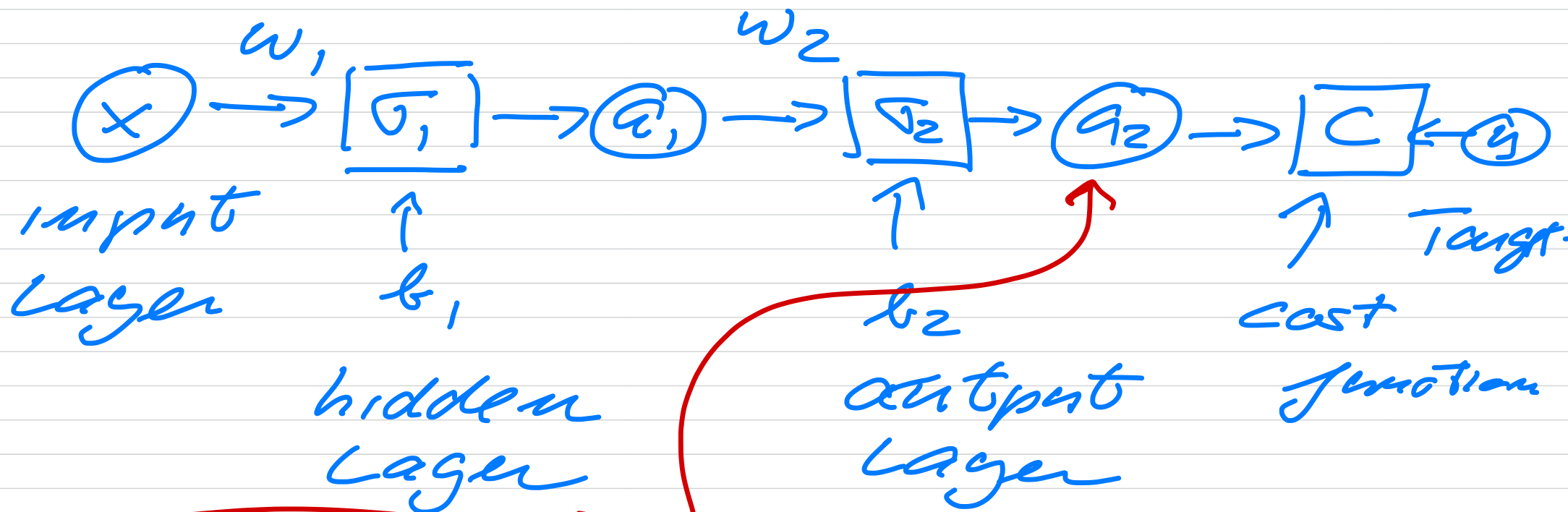


FYS-STK3155/4155 October 5



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From last week



$$\sigma_1 = \sigma_1(z_1) \quad \wedge \quad z_1 = w_1 x + b_1$$
$$\sigma_2 = \sigma_2(z_2) \quad \wedge \quad z_2 = w_2 a_1 + b_2$$

$$C = \frac{1}{2} (a_2 - y)^2$$

$$E = \{ w_1, w_2, b_1, b_2 \}$$

$$\frac{\partial C}{\partial E} = 0 \quad (a_2 - y) \quad \nabla_2 \quad \underline{a_1}$$

$$\frac{\partial C}{\partial w_2} = \frac{\partial C}{\partial a_2} \frac{\partial a_2}{\partial z_2} \frac{\partial z_2}{\partial w_2}$$

$$z_2 = w_2 a_1 + b_2 \quad \wedge \quad a_2 = \nabla_2(z_2)$$

$$\delta_2 =$$

$$\frac{\frac{\partial C}{\partial a_2} \frac{\partial a_2}{\partial z_2}}{(a_2 - y) \nabla_2'}$$

$$(a_2 - y) \nabla_2'$$

$$\frac{\partial C}{\partial w_2} = \delta_2 \cdot a_1 = \frac{\partial C}{\partial z_2} \frac{\partial z_2}{\partial w_2}$$

$$\frac{\partial C}{\partial b_2} = \frac{\partial C}{\partial a_2} \frac{\partial a_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial b_2} = \delta_2 \cdot \underline{1}$$

update rule

$$w_2 \leftarrow w_2 - \eta \delta_2 a, \quad b_2 \leftarrow b_2 - \eta \delta_2$$

$$\frac{\partial C}{\partial w_1} = \underbrace{\frac{\partial C}{\partial a_2} \frac{\partial a_2}{\partial z_2} \frac{\partial z_2}{\partial a_1} \frac{\partial a_1}{\partial z_1} \frac{\partial z_1}{\partial w_1}}_{\delta_2 \quad w_1 \quad \sigma_1' \quad x}$$

$$z_1 = w_1 x + b_1$$

$$a_1 = \sigma_1(z_1)$$

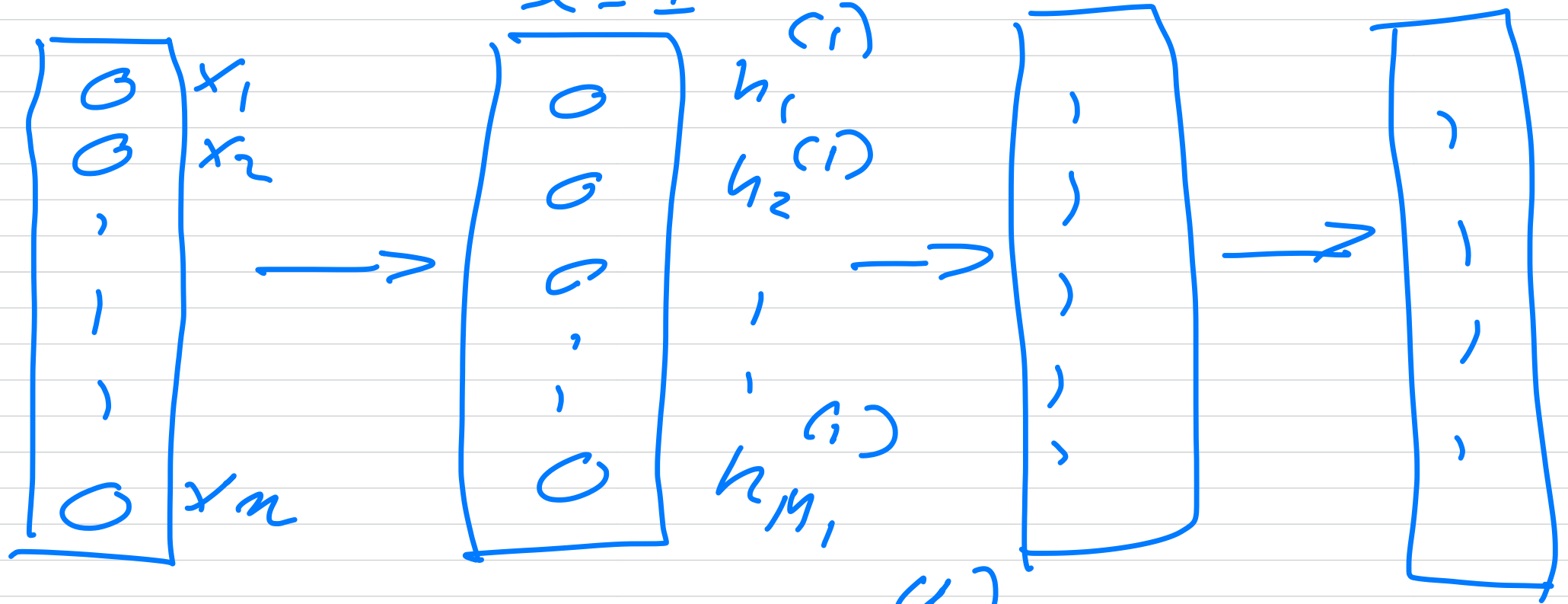
$$\frac{\partial C}{\partial z_1} = \delta_1 = \delta_2 w_1 \sigma_1'$$

$$w_1 \leftarrow w_1 - \eta \delta_1 x = w_1 - \eta \delta_1 a_0$$

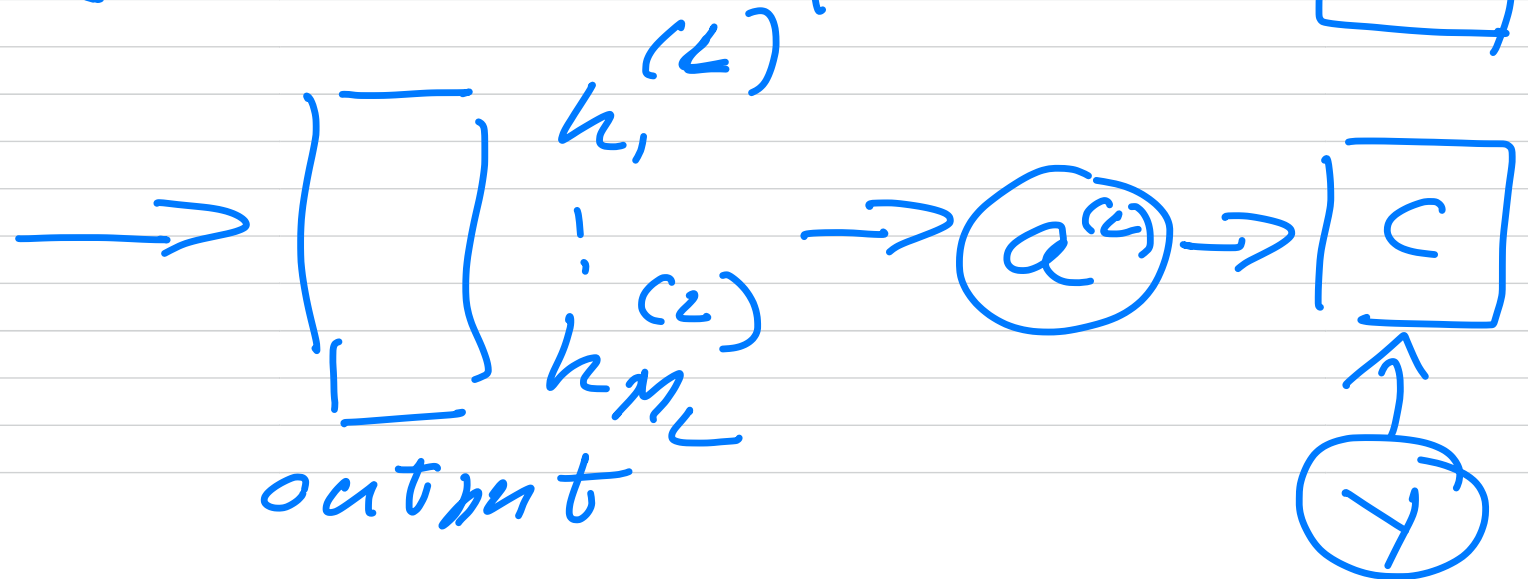
$$b_1 \leftarrow b_1 - \eta \delta_1$$

Back propagation algo

$l=1$

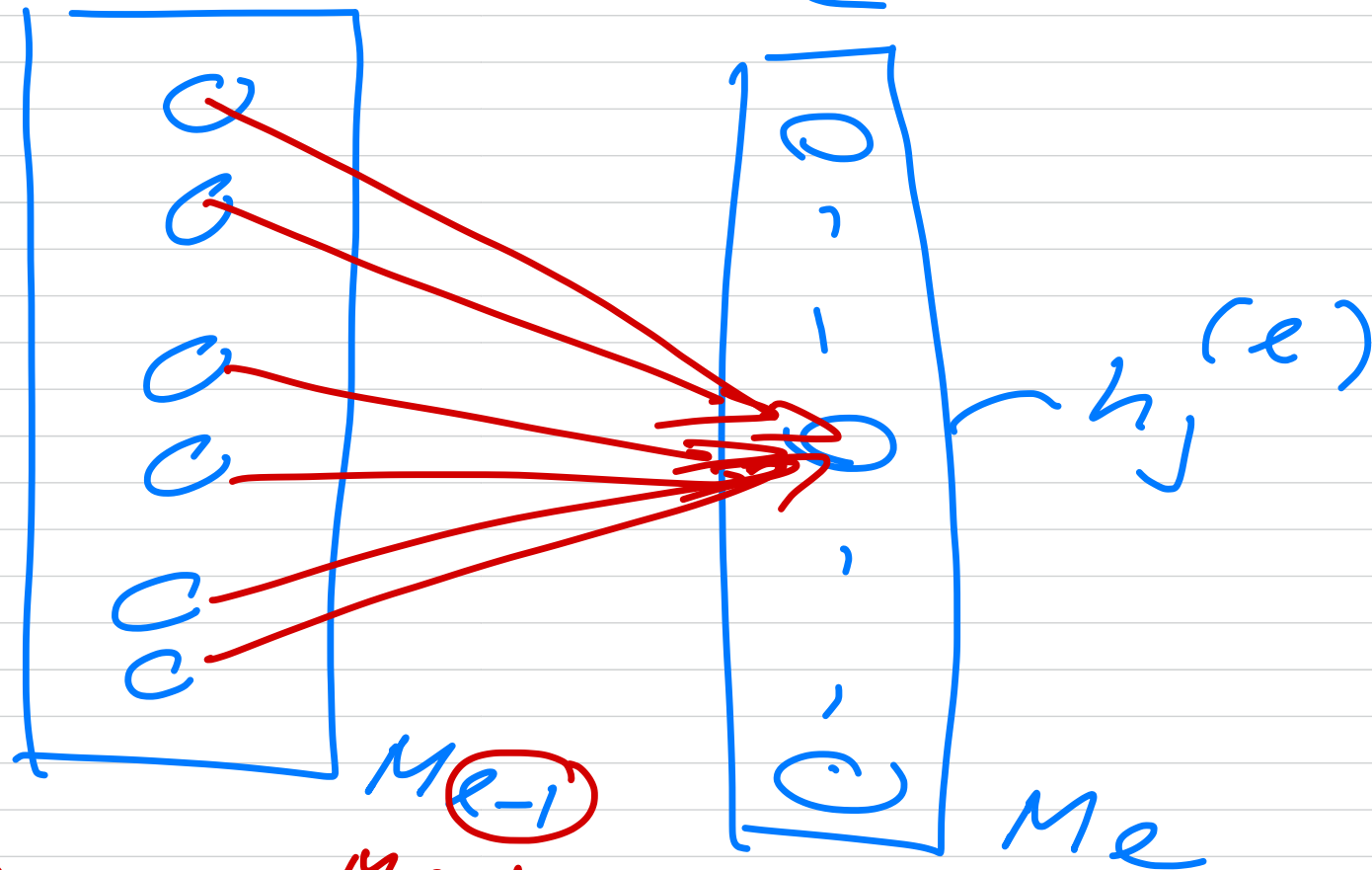


input
layer



output

one specific node in
layer l



$$(1) \quad z_j^{(l)} = \sum_{i=1}^{M_{l-1}} w_{ij}^{(l)} a_i^{(l-1)} + b_j^{(l)}$$

(ii) output from node $h_j^{(e)}$

$$a_j^{(e)} = \Delta(z_j^{(e)})$$

$$\vec{z}^{(e)} = (\vec{w}^{(e)})^T \vec{a}^{(e-1)} + \vec{b}^{(e)}$$

start with output layer

$$l = L$$

$$C(e) = \frac{1}{2} \sum_{i=1}^{M_L} (a_i^{(L)} - y_i)^2$$

$$E = \{w^{(1)}, w^{(2)}, \dots, w^{(L)}, b^{(1)}, \dots, b^{(L)}\}$$

$$\frac{\partial C}{\partial E^{(L)}} = 0$$

intermediate steps

$$- \frac{\partial z_j^{(L)}}{\partial w_{i'j}^{(L)}} = a_{i'}^{(L-1)}$$

$$- \frac{\partial z_j^{(L)}}{\partial a_i^{(L-1)}} = w_{i'j}^{(L)}$$

$$z_j^{(L)} = \sum_i w_{ij}^{(L)} a_i^{(L-1)} + b_j^{(L)}$$

Example $\sigma(z) = \frac{1}{1 + e^{-z}}$

$$\frac{d\sigma}{dz} = \underbrace{\sigma(z_j^{(l)}) (1 - \sigma(z_j^{(l)}))}_{a_j^{(l)} (1 - a_j^{(l)})}$$

$l = L$ (output)

$$\frac{\partial C}{\partial w_{jk}^{(L)}} = \underbrace{\frac{\partial C}{\partial a_j^{(L)}}}_{a_j^{(L)} - y_j} \underbrace{\frac{\partial a_j^{(L)}}{\partial z_j^{(L)}}}_{\sigma'} \underbrace{\frac{\partial z_j^{(L)}}{\partial w_{jk}^{(L)}}}_{a_k^{(L-1)}}$$

$$\delta_j^{(L)} = \underbrace{a_j^{(L)} (1 - a_j^{(L)})}_{\Delta_j} \underbrace{(a_j^{(L)} - y_j)}_{\frac{\partial \mathcal{L}}{\partial a_j^{(L)}}}$$

$$\frac{\partial \mathcal{L}}{\partial w_{jk}^{(L)}} = \delta_j^{(L)} a_k^{(L-1)}$$

$$(1) \quad w_{jk}^{(L)} \leftarrow w_{jk}^{(L)} - \eta \delta_j^{(L)} a_k^{(L-1)}$$

$$(2) \quad b_j^{(L)} \leftarrow b_j^{(L)} - \eta \delta_j^{(L)}$$

$$L \rightarrow l$$

$$\delta_j^{(l)} = \frac{\partial C}{\partial a_j^{(l)}} \frac{\partial a_j^{(l)}}{\partial z_j^{(l)}} = \frac{\partial C}{\partial z_j^{(l)}}$$

$$\delta_j^{(l)} = \frac{\partial C}{\partial z_j^{(l)}}$$

$$= \sum_k \frac{\partial C}{\partial z_k^{(l+1)}} \frac{\partial z_k^{(l+1)}}{\partial z_j^{(l)}}$$

$$z_j^{(l+1)} = \sum_{i=1}^M w_{ij}^{(l+1)} g_i^{(l)} + b_j^{(l+1)}$$

$$g_i^{(l)} = \Delta(z_i^{(l)}) = >$$

$$\frac{\partial z_k^{(l+1)}}{\partial z_j^{(l)}} = w_{kj}^{(l+1)} \cdot \Delta'(z_j^{(l)})$$

$$\delta_j^{(l)} = \sum_k \delta_k^{(l+1)} w_{kj}^{(l+1)} \Delta'(z_j^{(l)})$$

General update rules

$$w_{ij}^{(l)} \leftarrow w_{ij}^{(l)} - \eta \delta_j^{(l)} a_i^{(l-1)}$$

$$b_j^{(l)} \leftarrow b_j^{(l)} - \eta \delta_j^{(l)}$$

Algorithm for NNS

- (1) Define architecture
 - # nodes
 - # hidden layers

- activation functions
- cost/loss functions
+ L2/L1 etc norms
with hyperparameters
↓

(ii) - initialize all $\{W, b\}$
- setup design matrix

X

- Define learning rate γ
and gradient methods
and additional hyperparameters

(iii) while iterations $\leq$ max iter
OR stopping
criterion not reached

a) Perform Feed-Forward pass (ex week 4) from input via $l=1$ to $l=L$ = output layer, compare $a_j^{(L)}$ with target and compute score function

$l)$ - compute $\delta^{(l)}$
 and all other $\delta_j^{(l)}$
 $l = L-1, L-2, \dots, 1$
 update
 $w_{jk}^{(l)} \leftarrow w_{jk}^{(l)} - \eta \delta_j^{(l)} a_k^{(l-1)}$
 $b_j^{(l)} \leftarrow b_j^{(l)} - \eta \delta_j^{(l)}$
 end while

ex
 week
 42
 Back
 propagate